
- **OFFLINE JOURNEY BLIND SPOT**

Physical stores operate with zero path visibility. Customer browsing paths, zone dwell times, and shelf engagements remain completely untracked.

- **BILLING QUEUE ABANDONMENT**

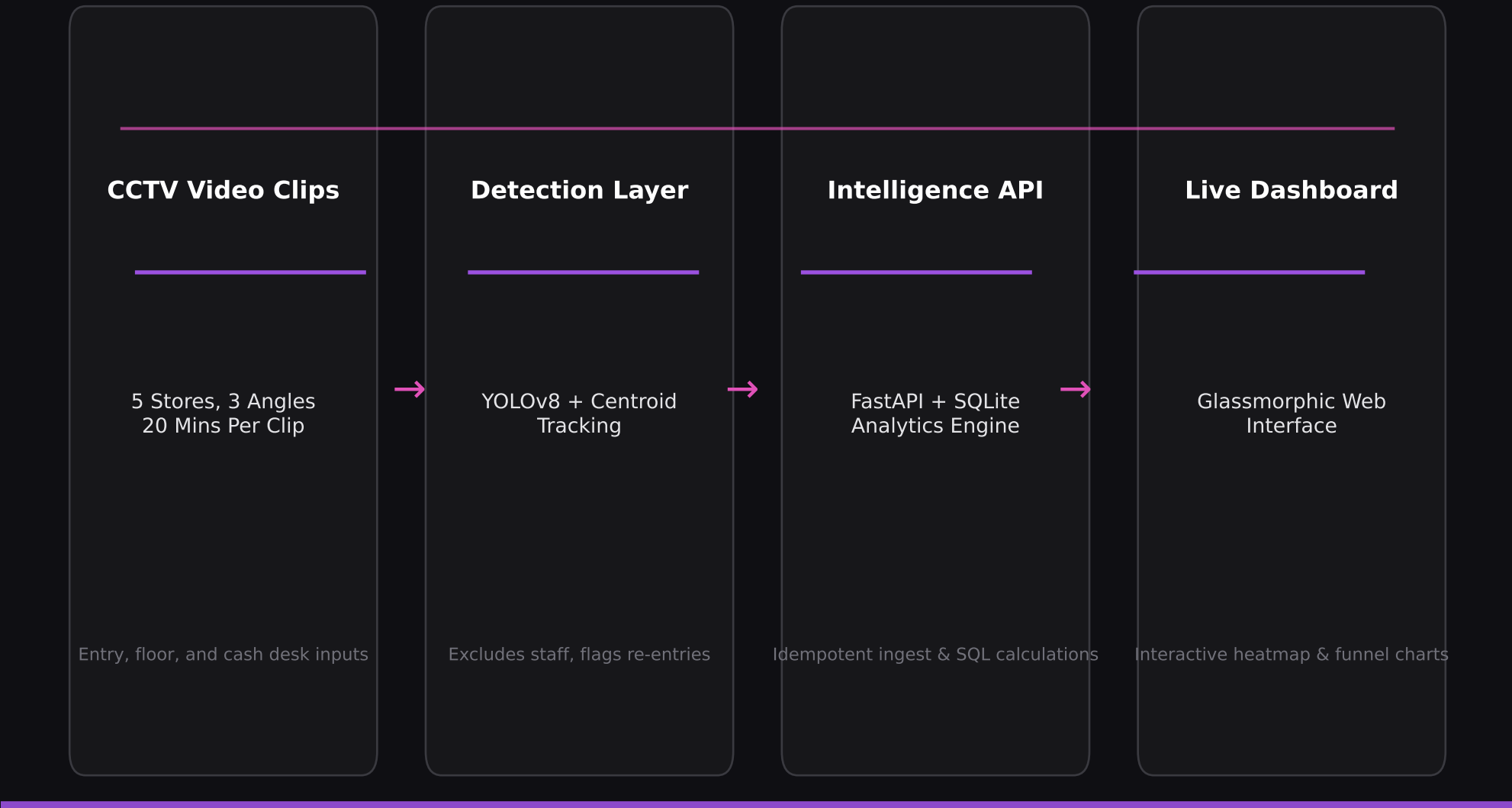
No analytical way to measure queue drop-offs. High abandonment rates at checkout go undetected due to a lack of queue monitoring.

- **DISCONNECTION FROM POS DATA**

Foot traffic is rarely correlated to transaction records. True checkout conversion rates cannot be calculated without time-window correlation.

- **HEAVY INFRASTRUCTURE OVERHEAD**

Traditional camera analytics require complex database servers and dedicated GPU hardware, blocking fast, edge-based deployment.



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- ✓ **Time-Window Conversion Correlation**
Correlates a visitor's checkout events to a POS transaction strictly if a transaction occurred within a 5-minute window following their presence. Resolves transaction-buyer mapping without customer identity data.
 - ✓ **Staff Exclusion Filter**
Excludes retail staff from customer conversion metrics using a color-histogram uniform classifier in the main floor tracking camera, tagging staff events with 'is_staff=true'.
 - ✓ **Idempotent Ingestion Safeguard**
Enforces event deduplication on API ingest using SQLite's native 'INSERT OR IGNORE' primary key index, ensuring safe, duplicate-free metrics updates on network retries.
 - ✓ **Self-Contained Edge Build**
Designed using serverless SQLite database, eliminating manual setup and service dependency issues on edge devices, allowing one-click Docker deployments.
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- 100% viewport-contained single-page layout (no scrolling)

- Dynamic Real-Time zone heatmap overlays (frequency scale 0-100)

- Active Conversion Funnel tracking (Entry -> Visit -> Queue -> Purchase)

- Live operational logs and active alerts for queues and dead zones

- Zero Setup Containerization (runs via 'docker compose up')

- Structured JSON logging tracing trace_id, store_id, and latency

- Graceful DB degradation mapping failures to HTTP 503 response

- High code coverage (>70% statement coverage in test suite)

Thank you! Live Demo at: purple-store-intelligence-xxm1.onrender.com
